Wenxiang Qin

Phone Number: (+86)137-7600-0046 E-mail: wenxiangqin341@gmail.com
Date of Birth: March.1994 Native Place: Suzhou, Jiangsu
Degree: Ph.D.

Education

Research Area: Robotic Vision 3D Vision Digital Twin

Apr.2022 — Mar.2026	Hiroshima University	Smart Robotics
Sep.2016 — Apr.2019	Nanjing University of Science and Technology	System Engineering
Sep.2012 — Jun.2016	Nanjing Agricultural University	Automation

Skill Tree

Theory

- Learned books: ”Matrix Theory”, ”Probability Theory”, ”Optimization”, ”Differential Geometry”, ”Statistical Learning Methods”, ”Pattern Recognition”, ”Machine Learning”, ”Deep Learning”, ”Fourteen Lectures on SLAM”, ”Principle of Automatic Control”, ”C++ Primer” and ”Introduction to Algorithms”

Math

- Familiar with mathematical tools such as advanced mathematics, matrix algebra, probability theory, differential geometry, linear and nonlinear optimization
- Good at using mathematics to analyze practical problems and design solutions

SLAM

- Familiar with structured-light-based 3D reconstruction methods
- Familiar with camera model, multi-view geometry, optimization, linear space; good at establishing and solving SLAM problems
- Familiar with vision-based SLAM methods, such as ORB-SLAM and DSO-SLAM, as well as their internal principles

Pattern Recognition

- Familiar with a variety of machine learning and statistical decision-making algorithms
- Familiar with the mathematical principles of deep learning

Basic Skills

- Familiar with C++ programming, skilled in algorithm development and library encapsulation
- Familiar with QT framework and its multi-threading mechanism, skilled in QT design and development
- Familiar with various data structures and CUDA parallel computing programming, skilled in writing high-performance algorithms
- Familiar with OpenCV, PCL, Eigen3, G2O, OpenGL, Sophus and other open source libraries
- Familiar with Git version management system, skilled in multi-feature parallel research and development
- Familiar with Microsoft Office, such as Word, Excel, Power Point, Visio, etc.
- Qualified in CET-6, with strong ability in English reading, writing, and dictation

Work Experience

Institute of Automation, Chinese Academy of Sciences **Technology R&D**

Structured-Light 3D Camera **Project Leader** **Jun.2018 – Aug.2020**

- Based on an integrated camera-projector system, high-speed 3D reconstruction of object surfaces (30 frames per second with 0.01 mm accuracy) is achieved using structured-light encoding/decoding and multi-view stereo geometry techniques. This technology has been successfully applied in multiple projects and in 3D camera products developed by the Suzhou Institute of Automation, Chinese Academy of Sciences.
  - Theoretical derivation of 3D spatial point reconstruction
  - Research and development of the structured-light-based 3D reconstruction algorithm
  - Research and development of coded pattern encoding and decoding algorithms
  - Research and development of the high-speed 3D reconstruction functionality
  - Architecture design and modular development of 3D cameras
  - Research and development of high-precision calibration methods for the projector–camera system

Universal Calibration System for Lens Models **Project Leader** **Jul.2020 – Mar.2022**

- Abstracting cameras and projectors as pinhole imaging models with individual distortions and poses, a universal camera-projector calibration framework was developed. This framework enables high-precision calibration for arbitrary number of cameras and projectors by jointly optimizing instrinsic and extrinsic parameters.
  - Modeling of the camera-projector system array
  - Design of the solving method for universal calibration
  - Design of data formats and software modular architecture

High-speed 3D Reconstruction and Point Cloud Analysis System for Moving Targets **Project Leader** **Mar.2020 – Oct.2020**

- Based on self-developed 3D cameras and structured-light cross-synchronized projection technology, real-time 3D modeling of live pigs is achieved using panoramic data acquired within 0.042 seconds. Combined with point cloud processing and multi-state joint estimation methods, pig point clouds were denoised, registered, segmented, and measured, enabling efficient and accurate extraction of multiple biometric parameters from live pigs.
  - Research and development of the multi-3D-camera joint calibration algorithm
  - Research and development of the multi-object joint ICP fine registration algorithm
  - Design of system structure
  - Design of software and algorithm modular architecture

3D Modeling System based on Single 3D Camera **Project Leader** **Apr.2019 – Mar.2022**

- Based on a single self-developed 3D camera, 3D object modeling was achieved by integrating image processing and pose estimation techniques. Loop-closure constraints were introduced using Lie algebra–based manifold modeling and optimization to effectively suppress accumulated errors. Computer graphics techniques were further applied for 3D model visualization and human-computer interaction.
  - Research and development of the 3D spatial manifold fitting algorithm
  - Research and development of the point cloud registration algorithm
  - Research and development of the loop-closure error elimination algorithm
  - Design of software and algorithm modular architecture
  - Development of the OpenGL 3D visualization module
  - Design and development of the 3D modeling system

ZHONG KE SHI JIE(Nanjing) Technology Co., Ltd CTO

Indoor Monitoring System for Earthquake Scenarios **Project Leader** **Oct.2020 – Dec.2020**

- Based on high-speed visual perception of patterned robotic markers, a 7-camera system covering full 360-degree scene was developed for 3D spatial coordinate estimation, enabling high-speed monitoring and recording of indoor object trajectories, vibrations, and oscillatory states.
  - Design of high-speed 3D perception scheme based on monocular camera and robotic markers
  - Design of software and algorithm modular architecture
  - Design of hardware configurations and mechanical structures
  - Team role allocation across software development, PC configuration, marker production, etc.

Human Body 3D Modeling System **Project Leader** **Sep.2020 – Mar.2022**

- Based on an array of self-developed 3D cameras and structured-light cross-synchronized projection technology, high-precision human 3D modeling was achieved.
  - Design of a full-body human modeling solution using 12 self-developed 3D cameras with cross-projection
  - Design of software and algorithm modular architecture
    (e.g. modules for capturing, 3D reconstruction, registration, calibration, and surface reconstruction)
  - Choice of camera, lens, and projector
  - Design of camera and projector angles, field-of-view coverage, and structural configuration

Alert System for Power Transmission Cables **Project Leader** **Jan.2021 – Jun.2021**

- A monocular-camera-based power transmission cable 3D reconstruction method was developed, and further enables real-time foreign object 3D detection by introducing another auxiliary camera, allowing assessing and recognizing potential threats against safe cable operation.
  - Research and development of the cable extraction algorithm
  - Research and development of the monocular-camera-based cable 3D reconstruction algorithm
  - Research and development of the stereo-measurement-based foreign object alerting algorithm

High-speed Dynamic Face 3D Reconstruction System **Project Leader** **May.2021 – Oct.2021**

- Based on three self-developed 3D cameras, high-speed and high-precision 3D modeling (200 models/s; 0.01 mm accuracy) of dynamic human faces were achieved, by integrating structured-light cross-synchronized projection technology, multi-3D-camera joint calibration algorithm, and multi-object joint ICP fine registration algorithm.
  - Design of system structure
  - Design of software and algorithm modular architecture
  - Construction and development of system

Ph.D. Research Topic

Digital Twin System for Vibrating Objects

- Targeting small vibrations of semi-rigid objects, a vision-based digital twin system was developed by integrating pre-defined 3D models with pre-assigned measurement points. This system enables vibration measurement and analysis, as well as their historical playback and comparison, with strong robustness against weak textures and high precision of 0.03 mm.
  - Research and development of the 3D-model-based vibration measurement and analysis algorithm
  - Design and construction of hardware system and mechanical structure
  - Experimentation, data organization and analysis, and paper writing

Camera pairing optimization for panoramic 3D vibration measurement

- For multi-camera-based panoramic 3D vibration measurement in the self-developed digital twin system, a camera pairing optimization method was developed by evaluating the reliability of individual cameras and their combinations for each measurement point. This camera pairing optimization method addresses background and self-occlusion interference problems, significantly improving measurement robustness in cluttered-background and complex-geometry scenarios.
  - Research and development of the camera evaluation and pairing optimization algorithm
  - Design and construction of hardware system and mechanical structure
  - Experimentation, data organization and analysis, and paper writing

Scientific Research Achievements

- **Wenxiang Qin**, Feiyue Wang, Shaopeng Hu, Kohei Shimasaki, Idaku Ishii, "Multi-Camera 3D Digital Image Correlation with Pointwise-Optimized Model-Based Stereo Pairing", *Sensors*, 2025, 25(18), Art. no. 5675.
- **Wenxiang Qin**, Feiyue Wang, Shaopeng Hu, Kohei Shimasaki, Idaku Ishii, "Model-Based 3D Vibration Measurement Using Stereo High-Frame-Rate Images", *IEEE Transactions on Instrumentation and Measurement*, 2025, 7(74), Art. no. 5040812.
- Feiyue Wang, **Wenxiang Qin**, Kohei Shimasaki, Idaku Ishii, Hiroshi Matsuda, "HFR-Video-Based 3-D Software Sensor for Bridge Displacement Monitoring", *IEEE Sensors Letters*, 2024, 8(10), pp. 1-4.
- **Wenxiang Qin**, Kohei Shimasaki, Feiyue Wang, Shaopeng Hu, Idaku Ishii, "Model Based Three-Dimensional Digital Image Vibration Measurement Method Using Stereo High-Speed Cameras", *14th International Conference of Optics-Photonics Design and Fabrication*, 2024(7).
- Tianwei Wu, Xilong Liu, Mengjuan Chen, Qingyi Gu, **Wenxiang Qin**, Tingyu Yan, Jiawei Lu, "Calibration and Measurement of Large Distortion Binocular Camera Based on Fully Connected Neural Network," *2021 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, Xining, China, July, 2021, pp. 833-838.

- **Wenxiang Qin**, Ling Guo, Shuhong Lin, "RGBD-based Box Size Measurement and Packaging Optimization", *Computer And Modernization*, 2019(5):46-50
- Idaku Ishii, Kohei Shimasaki, feiyue Wang, **Wenxiang Qin**, "Image analysis device, image analysis method and program", Invention, Patent No: JP2023046250
- Fan He, **Wenxiang Qin**, Qingqin Fu, Jie Ren, Yingchun Fu, Mengjuan Chen, "Automated Monitoring Method, System, and Apparatus for Relay Protection Hardwired Panels", Invention, Patent No: CN202210579484.4
- **Wenxiang Qin**, Qingyi Gu, Mengjuan Chen, "Method and System for Obtaining Spatial 3D Information of Overhead Transmission Lines based on Monocular Camera", Invention, Patent No: CN202210083995.7
- Xilong Liu, **Wenxiang Qin**, Qingyi Gu, Mengjuan Chen, "Method, System and Device for 3D Reconstruction Acceleration not Dependent on Mechanical Placement", Invention, Patent No: CN202110368059.6
- Xilong Liu, **Wenxiang Qin**, Qingyi Gu, Mengjuan Chen, "Method and System for Removing Strong Noise from Point Cloud of Complex Curved Surface", Invention, Patent No: CN201911098448.0
- **Wenxiang Qin**, Ling Guo, "RealSense-based Size Measurement System and Method", Invention, Patent No: CN201811400371.3
- Xilong Liu, Qingyi Gu, Mengjuan Chen, **Wenxiang Qin**, "Method, Device, Computer Equipment and Storage Medium for Measuring Spatial Straight Line Orientation", Invention, Patent No: CN201811244687.8

Individual Honors

- National Scholarship.
- The Japan Science and Technology Agency (JST) Research Fellowship, Grant Number JPMJFS2129.
- The Japan Science and Technology Agency (JST) "Support for Pioneering Research Initiated by the Next Generation (SPRING)", Grant Number JPMJSP2132.

Personal Evaluation

- Possess a solid foundation of basic knowledge, enabling effective handling of a wide range of related problems, with strong ability in integrating and applying knowledge across disciplines.
- Adept at identifying the underlying essence of problems and proposing practical and effective solutions, with a strong interest in cutting-edge technologies.
- Positive and optimistic, strong sense of responsibility, patient, and diligent in executing assigned tasks.
- Emphasizing on documentation, including formal mathematical recording of developed algorithms, systematic library integration, and a focus on modularity, independence, and reusability.
- Good at self-managing with the ability to reflect, synthesize knowledge frameworks, and identify clear directions for continuous progress.